



**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
**Off-Campus Centre of Nitte (Deemed to be University)**  
**Third Semester B.Tech (CBCS) Degree Examinations**  
 December 2024

**BT2103-1 – MOLECULAR BIOLOGY AND GENETIC ENGINEERING**  
 (For BT)

Duration: 3 Hours

Max. Marks: 100

**Note:**

**Part – A: Multiple Choice Questions:** Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B: Descriptive Answer Questions:** Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

**PART - A: MULTIPLE CHOICE QUESTIONS**

**20 Marks**

- What does the central dogma of molecular biology describe?**
  - Flow of information from DNA to RNA to Protein
  - The process of evolution
  - Synthesis of RNA from DNA
  - Synthesis of Protein from RNA
- Which classical experiment provided evidence that DNA is the genetic material?**
  - Avery McCleod experiment
  - Griffith's experiment
  - Hershey & Chase experiment
  - Gene cloning
- Which type of chromatin is typically associated with active transcription?**
  - Heterochromatin
  - Nucleosome
  - Metacentric
  - Euchromatin
- What is the primary function of telomeres in chromosomes?**
  - Cell division
  - To protect the ends of chromosomes from deterioration
  - Chromosomal separation
  - Meiotic division
- Which antibiotic inhibits translation by binding to the prokaryotic 30S ribosomal subunit?**
  - Rifampicin
  - Actinomycin D
  - Chloramphenicol
  - Tetracycline
- What was the key finding from the Hershey-Chase experiment regarding the role of proteins and DNA?**
  - DNA is the genetic material of the virus.
  - RNA is the genetic material of the virus.
  - Both DNA and proteins are responsible for carrying genetic information.
  - Proteins are the genetic material of the virus.
- During which process is the genetic code from mRNA used to build a protein?**
  - Transcription
  - Translation
  - Transversion
  - Transition
- What is the role of tRNA in the process of translation?**
  - To replicate DNA
  - To carry amino acids to the ribosome
  - To degrade mRNA
  - To transcribe DNA into mRNA
- Which of the following is NOT a standard classification based on the position of the centromere on chromosomes?**
  - Submetacentric
  - Metacentric
  - Telocentric
  - Polycentric
- During the elongation phase of protein synthesis, which site on the ribosome does the incoming aminoacyl-tRNA bind to?**
  - E site
  - P site
  - A site
  - T site
- During which phase of the cell cycle are chromosomes best visualized for karyotyping?**
  - Prophase
  - Cytokinesis
  - Metaphase
  - Meiosis - I

12. How does the complexity of a genome generally correlate with the number of protein-coding genes?
- A) More complex genomes have fewer protein-coding genes. B) More complex genomes have the same number of protein-coding genes.  
C) Genome complexity is not related to the number of protein-coding genes. D) More complex genomes have more protein-coding genes.
13. In which direction does DNA polymerase synthesize the new DNA strand?
- A) 3' to 5' B) 5' to 5'  
C) 3' to 3' D) 5' to 3'
14. Which enzyme removes RNA primers and replaces them with DNA nucleotides?
- A) DNA ligase B) DNA polymerase III  
C) DNA polymerase I D) Topoisomerase
15. Which process is involved in post-transcriptional modification of eukaryotic pre-mRNA?
- A) Addition of a poly(A) tail B) Addition of a 5' cap  
C) Splicing to remove introns D) All from the list
16. In the lac operon of E. coli, what molecule acts as an inducer to initiate the expression of the lac genes?
- A) Glucose B) Tryptophan  
C) ATP D) Lactose
17. Which enzyme is commonly used to join DNA fragments in the creation of recombinant DNA molecules?
- A) Nuclease B) DNA Ligase  
C) Terminal transferase D) Phosphatase
18. Which of the following best describes plasmids?
- A) Circular RNA molecules B) Circular DNA molecules found in bacteria  
C) Linear DNA molecules D) Linear RNA molecules found in viruses
19. What is the principle behind electroporation as a gene transfer technique?
- A) Using chemicals to disrupt cell membranes B) Using high voltage pulses to create pores in cell membranes  
C) Using viruses to deliver genetic material D) Using ultrasonic waves to enhance DNA uptake
20. Which of the following statements about liposome-mediated gene transfer is correct?
- A) It uses bacterial vectors for DNA delivery B) It involves direct DNA injection into cells  
C) DNA transfer is achieved by viral infection D) DNA is encapsulated within lipid vesicles for transfer into cells

### **PART - B: DESCRIPTIVE ANSWER QUESTIONS**

| Unit – I  |                                                                                                                                                                            | Marks | BT* | CO* | PO*  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|------|
| 1.        | a) Discuss the classical experiments that led to the identification of DNA as the genetic material.                                                                        | 8     | L*1 | 1   | 2,12 |
|           | b) Write short note on: i) Central dogma; ii) DNA Polymerases                                                                                                              | 8     | L1  | 1   | 2,12 |
| 2.        | a) What is a replication fork? With a neat, labelled diagram, provide the details of various components present in the replication fork and their functional significance. | 8     | L1  | 1   | 2,12 |
|           | b) What is DNA damage? List out various types of DNA damages. Discuss the mechanism of Base Excision Repair.                                                               | 8     | L2  | 2   | 2,12 |
| 3.        | a) Discuss in detail the events of reverse transcription process.                                                                                                          | 8     | L1  | 2   | 2,12 |
|           | b) With respect to Chromosomes, discuss the following: i) Molecular organization, ii) Classification                                                                       | 8     | L1  | 2   | 2,12 |
| Unit – II |                                                                                                                                                                            |       |     |     |      |
| 4.        | a) Elaborate the structure and functions of RNA polymerase. Add a note on its significance.                                                                                | 8     | L2  | 3   | 6,12 |
|           | b) List the similarities and differences between prokaryotic and eukaryotic RNA polymerases.                                                                               | 8     | L1  | 3   | 6,12 |

- |       |                                                                               |     |    |   |      |
|-------|-------------------------------------------------------------------------------|-----|----|---|------|
| 5. a) | With appropriate illustrations, discuss the general mechanism of translation. | 8   | L2 | 3 | 6,12 |
| b)    | Write short notes on: i) Post-transcriptional modification, ii) Genetic code. | 4+4 | L1 | 4 | 2,12 |
| 6. a) | With suitable example, discuss prokaryotic gene regulation process.           | 8   | L2 | 4 | 6,12 |
| b)    | Give a detailed account of prokaryotic transcription process.                 | 8   | L1 | 4 | 2,12 |

**Unit – III**

- |       |                                                                                                                                    |   |    |   |      |
|-------|------------------------------------------------------------------------------------------------------------------------------------|---|----|---|------|
| 7. a) | Discuss the physical modes of gene transfer techniques used in genetic engineering.                                                | 8 | L1 | 5 | 2,12 |
| b)    | Outline the steps involved in creating recombinant DNA molecules.                                                                  | 8 | L1 | 5 | 2,12 |
| 8. a) | Elaborate the unique properties of Agrobacterium's Ti and Ri plasmids that make them effective tools in plant genetic engineering. | 8 | L1 | 5 | 2,12 |
| b)    | Discuss in detail about DNA modifying enzymes.                                                                                     | 8 | L1 | 5 | 2,12 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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